



AEO "Nazarbayev Intellectual Schools"

CHEMISTRY

Grade 12

Paper 2

May 2020

MARK SCHEME

Maximum Mark: 100

This document consists of **9** printed and **3** blank pages.

PRE-STANDARDISATION

Question	Answer	Mark	Additional guidance
1 (a) (i)	$\text{Cu} + 2\text{H}_2\text{SO}_4 \rightarrow \text{CuSO}_4 + \text{SO}_2 + 2\text{H}_2\text{O}$	1 1 [2]	Correct formulae Balancing – dependent on correct formulae
1 (a) (ii)	Amount of substance that contains Avogadro's number of specified particles	[1]	Accept The relative atomic mass/relative formula mass in grams of any element/compound $6,02 \times 10^{23}$ molecules/ions/atoms amount of substance in 12 grams of carbon-12
1 (a) (iii)	$192 / 63.5 = 3.0$ moles Cu 1:1 (from the equation of reaction) 3 moles Cu^{2+} and 6 moles NO_3^-	1 1 1 [3]	Accept answer 3.0 - 3.02 moles; 6.0 – 6.02 moles ecf for third marking point from moles of copper e.g. if moles of Cu=2 then answer should be 2 moles Cu^{2+} and 4 moles NO_3^-
1 (b)	$\text{MnO}_4^{2-} + \text{H}_2\text{O} \rightarrow \text{MnO}_2 + \text{MnO}_4^- + \text{OH}^-$ $3\text{MnO}_4^{2-}(\text{aq}) + 2\text{H}_2\text{O}(\text{l}) \rightarrow \text{MnO}_2(\text{s}) + 2\text{MnO}_4^-(\text{aq}) + 4\text{OH}^-(\text{aq})$	1 1 [2]	Note one mark for the correct ionic species in the equation even if K^+ is shown one mark for the balanced ionic equation with state symbols
1 (c) (i)	(heat changes when) <u>1 mole of gaseous ions</u> is dissolved to form dilute solution/ in lots of water.	[1]	
1 (c) (ii)	$\text{K}^+(\text{g}) \rightarrow \text{K}^+(\text{aq})$ $\text{MnO}_4^-(\text{g}) \rightarrow \text{MnO}_4^-(\text{aq})$	1 1 [2]	Accept $\text{K}^+(\text{g}) + \text{aq} \rightarrow \text{K}^+(\text{aq})$ $\text{MnO}_4^-(\text{g}) + \text{aq} \rightarrow \text{MnO}_4^-(\text{aq})$

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Question	Answer	Mark	Additional guidance
1 (c) (iii)		[3]	Note 1 mark for lattice enthalpy change and correct direction arrow 1 mark for change of enthalpy of hydration correct direction arrow 1 mark for change of enthalpy of solution correct direction arrow
2 (a)	Sulfur found in nature as brimstone/ /coal/crude oil/volcanoes	[1]	Accept minerals and salts
2 (b) (i)	Being the same element, it has different forms in the same state	[1]	Accept Different crystal or molecular lattices of the same element
2 (b) (ii)	<i>Similarity:</i> they are both S ₈ / both have eight sulfur atoms in the molecule/ they're both yellow in colour, have single bonds and are in a ring. <i>Difference:</i> Monoclinic Sulphur is unstable at temperatures below 96°C/ β-sulfur, non-stable, the puckered ring/ prismatic/ bond angle	1 [2]	Accept Must have a comment about both rhombic and monoclinic sulfur
2 (c) (i)	1. 2SO ₂ (g) + O ₂ (g) ⇌ 2SO ₃ (g) 2. SO ₃ (g) + H ₂ SO ₄ (l) → H ₂ S ₂ O ₇ (l) 3. H ₂ S ₂ O ₇ (l) + H ₂ O(l) → 2H ₂ SO ₄ (l) / H ₂ S ₂ O ₇ (l) + H ₂ O(l) → 2H ₂ SO ₄ (aq)	1 1 1 [3]	Do not accept unbalanced equations Accept Reaction arrow in step 1 2 marks for 3 correct reactions without state symbols Correct multiples including fractions e.g. SO ₂ (g) + 1/2O ₂ (g) ⇌ SO ₃ (g)
2 (c)(ii)	to increase the rate of reaction without shifting an equilibrium too far to the left pressure high enough to have the position of the equilibrium on the right/product to shift the position of an equilibrium to the right	1 1 1 [3]	Accept owtte owtte owtte

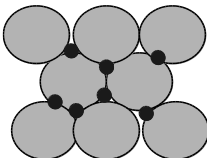
Question	Answer	Mark	Additional guidance
2 (d)	0.10 moles (H_2SO_4) 0.30 moles (NaOH) (after the addition) 0.1 mole NaOH $(0.1/0.4) / 0.25 \text{ mol dm}^{-3} \text{ NaOH}$ pH = 13.4	1 1 1 1 1 [5]	Accept ecf for third marking point e.g. moles NaOH -2x moles H_2SO_4 ecf for fourth marking point e.g. moles after addition/0.4 ecf for fifth marking point answer 13.0-13.5
3 (a)	Proteins, viruses, DNA	[1]	1 mark for all three answer
3 (b) (i)	Graphene OR Buckminsterfullerene / fullerene Graphene has a hexahedron/hexagonal lattice of one layer carbon atoms. OR Buckminsterfullerene has a spherical shape with pentagonal and/or hexagonal panels.	1 1 [2]	Accept buckyball Note The name and its description should match for the second marking point
3 (b) (ii)	good Adsorption/large surface area	1 [1]	Ignore small size
3 (c) (i)	Any three from: An idea that drugs are located inside nanoparticles An idea that polymer chains are bonded with nanoparticles An idea that pores of cell are opened on heating Nanoparticles enter the cell and release drug Nanoparticle then breaks down	1 1 1 1 1 [max3]	Accept Correct diagram/picture with drugs and polymer owtte

Question	Answer	Mark	Additional guidance
3 (c) (ii)	Any three from: An idea that less side effects PEG hydrolysis/ transports only two molecules of drug An idea that nanoparticles protect the drugs from digestive system/hydrolysis reaction/ acid Can selectively target cells The molecules can be effectively removed by kidney Small enough to go through the cell membrane Can be made soluble Polymer coating reduces poisonous nature of drugs Nanoparticles may be poisonous	1 1 1 1 1 1 1 1 1 [max3]	Accept expensive/high cost Accept answer 3 (c) (i)
4 (a)	For correct use formula/ $([3 \times -393.5]) - (-824.2 + [3 \times -110.5])$ For correct answer $-24.8 \text{ (kJ} \times \text{mol}^{-1}\text{)}$ For correct use formula/ $([2 \times 27.3] + [3 \times 213.6]) - (87.4 + [3 \times 197.6])$ For correct answer $(+) 15.2 \text{ (J K}^{-1} \text{mol}^{-1}\text{)}$	1 1 1 1 [4]	
4 (b)	For correct formula For correct answer 908 K with 3 significant figures	1 1 [2]	
4 (c)	ΔH^θ for carbon reaction is more than for carbon monoxide/ more endothermic reaction (need high temperature)/reaction for carbon monoxide is more exothermic ΔS^θ for carbon reaction is larger than for carbon monoxide	1 1 [2]	Accept as signs > and < Enthalpy contribution in reaction with carbon monoxide is more Entropy contribution in reaction with carbon monoxide is less

Question	Answer	Mark	Additional guidance
4 (d)(i)	Transport (molecular) oxygen (in blood)	[1]	owtte
4 (d)(ii)	idea that (the dative) bond to CO is stronger than that to O ₂ presence of (dative) bond of the CO reduces ability to bond with O ₂	1 1 [2]	Accept Coordinated bond stronger/polar Need more energy to break between CO and haemoglobin
5 (a) (i)	G is an alpha helix K is a beta pleated sheet / β -sheet	1 1 [2]	
5 (a) (ii)	Hydrogen bonding Between amino acids the H in the N-H bond of the peptide link and the O in the C=O bond of the peptide link	1 1 [2]	
5 (a) (iii)	The primary structure of a protein is the <u>sequence of amino acid</u> residues in the polypeptide chain that makes up the protein molecule. – NH – CO – Peptide bond / amide linkage between amino acids	1 1 1 [3]	Ignore covalent bond
5 (b)	Change of structure/properties of the protein (due to external factors)	[1]	
5 (c)	Increase temperature / heating cooking of egg / albumin goes solid and/or changes colour change of pH / add alkali / add acid casein in milk goes solid	 [2]	Note Two correct methods = 1 mark Two correct examples = 1 mark Correct method and example = 1 mark Accept Any suitable examples
5 (d)	Protein is precipitated bonding/structure/shape decomposes/changes	1 1 [2]	Accept If answer is shown as equations $--CH_2-S-S-CH_2-- + Cu^{2+} \rightarrow 2 (-CH_2-S^-) Cu^{2+}$ Solubility of the protein is decreased

Question	Answer	Mark	Additional guidance
6 (a) (i)	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^5 / [Ar] 3d^5$	[1]	Ignore $4s^0$
6 (a) (ii)	$[Fe(H_2O)_6]^{3+}$ $[Fe(H_2O)_6]^{3+}$ and Fe^{3+} has a greater charge density / smaller ion / higher charge so more polarisation of the O-H bond in the water (ligand) / more distortion of the shared pair of electrons in O-H bond	0 1 1 [2]	Accept More polarizing power O-H bond in the water getting weaker produce more H^+ Note if $[Fe(H_2O)_6]^{2+}$ given as answer no marks for the question
6 (b) (i)	$[Fe(H_2O)_6]^{2+}(aq) + CO_3^{2-}(aq) \rightarrow FeCO_3(s) + 6H_2O(l)$ balanced equation correct state symbols – dependent on correct formulae	1 1 [2]	
6 (b) (ii)	(light) green precipitate	[1]	Accept Any shades of green
6 (c) (i)	$2[Fe(H_2O)_6]^{3+} + 3CO_3^{2-} \rightarrow 2[Fe(H_2O)_3(OH)_3] + 3CO_2 + 3H_2O$	[1]	Ignore State symbols
6 (c) (ii)	red-brown precipitate/bubbles of gas	[1]	Accept Brown Ignore Gas formed or carbon dioxide gas
6 (d)	$K_{stab} = \frac{[Fe(CN)_6]^{4-}}{[Fe(H_2O)_6]^{2+} [CN^-]^6}$	[1]	
6 (e) (i)	Chelate effect	[1]	Accept Description of chelate effect
6 (e) (ii)	Any three from Entropy change is positive/products have more entropy Because more particles in product Enthalpy change is negligible Gibbs free energy change (large) and negative	1 1 1 1 [max 3]	Ignore Enthalpy change is exothermic

Question	Answer	Mark	Additional guidance												
7 (a) (i)	Any three from Butan-1-ol $\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-CH}_2\text{OH}$ Butan-2-ol $\text{CH}_3\text{-CH(OH)-CH}_2\text{-CH}_3$ 2-methylpropan-1-ol $\text{CH}_3\text{-CH(CH}_3\text{)-CH}_2\text{OH}$ 2-methylpropan-2-ol $\text{CH}_3\text{-C(CH}_3\text{)(OH)-CH}_3$	1 1 1 1 [max 3]	The names to corresponds the structural formulae Accept $(\text{CH}_3)_3\text{COH}$ without showing the bonds e.g. $\text{CH}_3\text{C(CH}_3\text{)(OH)CH}_3$ 1 mark for all correct structures without names												
7 (a) (ii)	elimination / dehydration	[1]													
7 (a) (iii)	$\text{H}^+/\text{H}_2\text{SO}_4 / \text{H}_3\text{PO}_4 / \text{Al}_2\text{O}_3 / \text{ZnCl}_2 / \text{SiO}_2$ Higher than $200^\circ\text{C} / 300^\circ\text{C} / 400^\circ\text{C}$	[1]	both answers needed Accept high temperature Ignore Heat												
7 (b) (i)	<table border="1"> <thead> <tr> <th></th><th>higher rate of reaction</th><th>more expensive equipment</th><th>lower atom economy</th></tr> </thead> <tbody> <tr> <td>fermentation</td><td></td><td></td><td>✓</td></tr> <tr> <td>hydration</td><td>✓</td><td>✓</td><td></td></tr> </tbody> </table>		higher rate of reaction	more expensive equipment	lower atom economy	fermentation			✓	hydration	✓	✓		[max 2]	One incorrect tick maximum 1 mark two incorrect ticks 0 marks three correct ticks 2 marks two correct ticks 1 mark
	higher rate of reaction	more expensive equipment	lower atom economy												
fermentation			✓												
hydration	✓	✓													
7 (b) (ii)	$6\text{CO}_2 + 6\text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$ $\text{C}_6\text{H}_{12}\text{O}_6 \rightarrow 2\text{C}_2\text{H}_5\text{OH} + 2\text{CO}_2$ $2\text{C}_2\text{H}_5\text{OH} + 6\text{O}_2 \rightarrow 4\text{CO}_2 + 6\text{H}_2\text{O}$ number of moles of carbon dioxide to make glucose equals number of moles of carbon dioxide made	2 1 [3]	Do not accept unbalanced equation Note 1 mark for one correct equation 2 marks for all three correct equations												
7(c) (i)	from orange to green	[1]	Accept blue green												
7 (c)(ii)	Ethanoic acid/acetic acid/ $\text{CH}_3\text{COOH} / \text{CH}_3\text{CO}_2\text{H}$	[1]													

Question	Answer	Mark	Additional guidance
8(a)(i)	Correct structure Aluminum ions Free electrons/delocalized electrons/sea of electrons	1 1 1 [3]	Correct diagram 
8(a) (ii)	It has crystal/metallic lattice/metallic bond Strong electrostatic attraction / strong attraction between cations and electrons	1 1 [2]	Accept Al^{3+} Al^{2+} Al^+ instead of cation
8(a) (iii)	heat transferred by mobile electrons/ kinetic energy of electrons	1 1 [2]	
8 (b)	$Al^{2+}(g) \rightarrow Al^{3+}(g) + e^-$	[2]	Note 1 mark for correct equation 1 mark for correct state symbols – dependent on correct formulae (1)
8 (c)	Dot-and-cross diagram showing: Lack/deficiency of electrons on the outer shell of aluminium Lone pair from two chlorine atoms forms dative/coordinate bonding with aluminium atoms	1 1 [2]	
8 (d)	Aluminium requires 5 electrons to complete its octet. Monomer ($AlCl_3$) is unstable due to unpaired electron in p-orbital of Al , dimer (Al_2Cl_6) so that it becomes stable.	1 1 [2]	Accept Dimer forms through a coordinate/dative covalent bond to complete the octet of aluminium using lone pairs of chlorine atoms.

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